Probing Warmth and Competence Representations in LLM Hiring Decisions

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Abstract

Discrimination in hiring has been documented in both humans and large language models, yet the internal mechanisms that produce it remain poorly understood. We probe whether the Stereotype Content Model dimensions of warmth and competence are encoded as linear directions in the residual streams of four open-weights instruction-tuned language models (Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, and Qwen3-14B), and whether those representations causally shift hiring-callback recommendations when directly steered. We find that warmth and competence are linearly separable with exceptional effect sizes across all four architectures, that the directions are causally functional in concept-level judgements, and that the Gemma family partially aligns with human social perception ratings while Llama and Qwen show warmth anti-alignment. The causal chain from representation to hiring decision is fragile and model-dependent: the most concept-steerable model shows no significant mediation path, and the larger Gemma model reverses the human racial callback gap by more than one standard deviation, consistent with instruction-tuning overcorrection rather than inherited discrimination.

Introduction

AI-powered tools now screen resumes, rank candidates, and support callback decisions at every stage of the modern hiring funnel. According to SHRM's 2025 Talent Trends survey, half of U.S. organizations use AI to support recruiting, and 44 percent deploy it specifically for resume screening (Society for Human Resource Management, 2025). At that scale, any systematic skew in model recommendations translates directly into skewed access to work.

That skew is not hypothetical. Decades of human correspondence studies have established measurable hiring discrimination across race, national origin, age, disability, and gender (Bertrand and Mullainathan, 2004; Oreopoulos, 2011; Neumark et al., 2019; Correll et al., 2007; Tilcsik, 2011; Ameri et al., 2018). Whether automating the hiring decision preserves or amplifies these patterns is now a live question. Several behavioral audits of large language models suggest it does: Wilson and Caliskan (2024) and Chaturvedi and Chaturvedi (2025) find that LLMs systematically favor some demographic groups over others in resume evaluation, while An et al. (2024), An et al. (2025), and Gaebler et al. (2024) document persistent race and gender disparities in GPT-family callback recommendations. These studies agree that bias exists; they disagree on its direction and magnitude. None can locate the internal mechanism: they describe what the model decides, not how it arrives there.

Mechanistic interpretability offers a way in. Word embedding research established that semantic concepts cluster as linear directions in high-dimensional activation spaces (Mikolov et al., 2013), and the linear representation hypothesis formalizes this for modern large models (Park et al., 2024). Representation engineering (Zou et al., 2023) and activation addition (Turner et al., 2023) showed that steering a model along such a direction causally shifts its behavior, and circuit analyses have connected individual components to recognizable computations (Olah et al., 2020). Most recently, Sofroniew et al. (2026) demonstrated that emotion concepts in Claude Sonnet are encoded as linear residual-stream directions and causally regulate behavior under direct steering—evidence that the geometry of representation extends from symbolic word vectors to the internal activations of instruction-tuned models. We borrow this technique and apply it to a different construct entirely.

The construct is warmth and competence—the two axes through which social groups are perceived according to the Stereotype Content Model (Fiske et al., 2002). Warmth and competence ratings of social signals predict

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human callback rates across decades of North American correspondence studies (Gallo et al., 2024), making these axes a natural candidate mechanism for LLM hiring disparities. We ask whether four open-weights models have internalized warmth and competence as linearly decodable residual-stream directions, whether those directions are causally functional, and whether they mediate the name-driven callback disparities we observe against a matched human benchmark. That question connects mechanistic interpretability to the empirical tradition of correspondence studies in a domain where the stakes of getting it wrong are immediate and measurable.

Background: Reading a Concept Out of a Language Model

Activations and the Residual Stream

Every forward pass through a transformer accumulates computation through additive residual connections: each layer ℓ reads a d_{model}-dimensional state $h_\ell \in \mathbb{R}^{d_{\text{model}}}$ and writes back an updated version $h_{\ell+1} = h_\ell + f_\ell(h_\ell)$, where f_ℓ is the combined output of that layer's attention and MLP sub-blocks. Circuit-level analyses have shown that specific features—from low-level syntax to high-level semantic content—are localized within this stream as consistent geometric patterns across contexts (Olah et al., 2020). Rather than probing the full architecture, we extract the residual-stream state at a single chosen depth (approximately 65% of network layers, selected by Cohen's d emergence curves) and treat that vector as the space in which warmth and competence reside as directions.

A Concept Is a Direction

The intuition that semantic content is organized as linear directions dates to word-embedding analogies (Mikolov et al., 2013); the *linear representation hypothesis* (Park et al., 2024) formalizes it for large transformer models: high-level features correspond to directions in residual-stream space such that projecting a representation onto a direction yields a scalar measuring how strongly that feature is present. To extract such a direction for a target concept, we collect residual-stream activations from texts that express the concept (condition A) and from matched texts that do not (condition B), and form the mean-difference:

$$v = \bar{h}_A - \bar{h}_B. \tag{1}$$

Normalizing to $\hat{v}$ gives a concept direction; the concept score of any new text is then $\hat{v}^\top h$. The key claim — validated here by cross-validated classification and topic-holdout accuracy — is that this direction generalizes beyond the specific texts used to construct it (Fig. 2).

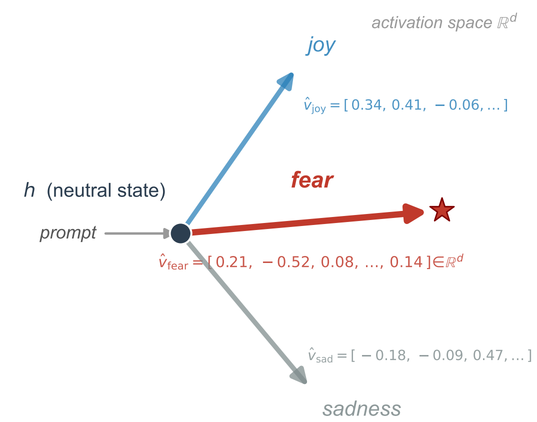

Figure 1: Emotion vectors as directions in activation space (schematic). Each emotion corresponds to a unit direction radiating from a neutral model state h. Feeding a prompt that evokes fear shifts the residual stream along $\hat{v}_{\text{fear}}$, encoded here with its representative d-dimensional coordinates. After Sofroniew et al. (2026); we reuse the geometry, not the construct.

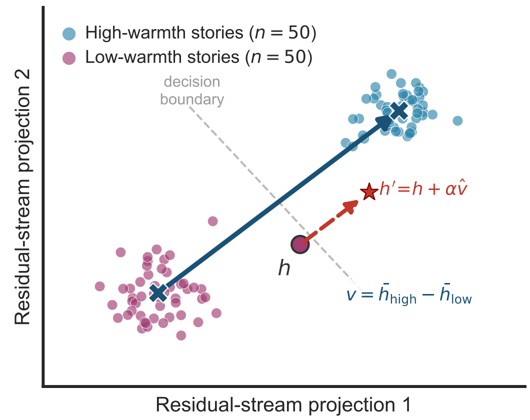

Figure 2: Residual-stream concept geometry (schematic). Two story clusters separated by the mean-difference direction $v = \bar{h}_{\text{high}} - \bar{h}_{\text{low}}$ (blue arrow). The red dashed arrow shows a single steering step $h' = h + \alpha\,\hat{v}$ that moves a marginal point across the decision boundary. Illustrative synthetic activations, not measured data.

Steering a Concept

A direction extracted by the mean-difference method is initially correlational: it separates high- and low-condition activations but does not by itself demonstrate that the model uses the direction causally. To test causality, we intervene on the residual stream during inference: at the probe layer, we replace the model's internal state with

$$h' = h + \alpha\,\hat{v}, \tag{2}$$

where α is a signed scalar controlling the magnitude and direction of the push. If the model's downstream behavior then shifts systematically with α while an orthogonal control direction produces no comparable shift, the direction participates causally in the model's computations. Activation addition (Turner et al., 2023) and representation engineering (Zou et al., 2023) establish this approach and give it empirical grounding across a range of models and target behaviors.

Emotion Vectors as the Template

Sofroniew et al. (2026) applied this framework to emotion concepts in Claude Sonnet. Building contrastive text pairs for each emotion, they extracted mean-difference directions in the residual stream, steered the model along those directions, and observed systematic, interpretable shifts in behavior. Their study also introduced the neutral-corpus PCA denoising procedure that removes general evaluative valence from the direction vectors—a step we adopt verbatim to separate warmth and competence content from shared narrative polarity (described in Methods). We reuse the geometry and the method; the construct is different. Warmth and competence are social-perception dimensions from the Stereotype Content Model, not emotion states, and calling our direction vectors "emotion vectors" would be a category error. Fig. 1 illustrates the core idea: each emotion occupies a distinct direction in residual-stream space, with a numeric representation that can be extracted, inspected, and used to steer the model.

Methods

Story Corpus. To construct concept direction vectors independently of demographic identity, we generated 200 short narratives (approximately 90–150 words each) distributed evenly across four conditions: high warmth, low warmth, high competence, and low competence (50 stories per condition). Stories were written around 50 distinct everyday topics spanning ten domains (workplace, learning, community, sport, travel, and others), with each topic appearing once per condition so that situational vocabulary is balanced across the high and low poles. All stories feature a nameless, gender-neutral third-person protagonist ("they"). This design choice was motivated by pilot work showing that AI-generated concept stories exhibit systematic demographic skew: low-warmth and low-competence conditions were overwhelmingly assigned male, Anglo-named protagonists, while high conditions were assigned female and demographically diverse protagonists. Without neutralisation, direction vectors would encode demographic expectations alongside the target social dimension, confounding the probe with the very bias we seek to study. By removing identity signals entirely, each story's activation is driven by depicted behaviour rather than group membership. Stories were generated by a large language model and validated by automated checks for forbidden-vocabulary violations, length balance across conditions, and demographic neutrality. The use of AI-generated stimuli raises concerns about circularity that we address in the limitations. The current corpus of 200 stories is a pilot; we plan to expand to 400 stories as data collection continues.

Table 1: Model parameters and probe extraction settings. Probe layer is given as selected/total layers. $\|\bar{h}\|$ is the mean residual-stream norm at the probe layer, which determines the scale of steering interventions; the orders-of-magnitude difference across models makes raw-strength comparisons uninformative without normalisation.

Model	d_{model}	Probe layer	$\|\bar{h}\|$
Gemma-3-12B-it	3840	31/48	79,722
Gemma-3-27B-it	5376	40/62	61,576
Llama-3.1-8B-Instruct	4096	20/32	11.4
Qwen3-14B	5120	26/40	206.6

Activation Extraction. We used TransformerLens (Nanda and Bloom, 2022) to extract residual-stream activations from four open-weights instruction-tuned models: Gemma-3-12B-it (Gemma Team, 2025) as the primary model, Gemma-3-27B-it as a within-family scale replication, and Qwen3-14B (Yang et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) as cross-architecture replications. For each model we captured the post-layer residual-stream state at the layer corresponding to approximately 65 per cent of model depth, selected on the basis of Cohen's d emergence curves from a full layer sweep. Tab. 1 reports the exact probe layer, model width, and mean residual-stream norm for each model. For each story, activations were mean-pooled across token positions from position 50 onward, skipping the prompt prefix, yielding one d_{model}-dimensional vector per story. Warmth and competence direction vectors were then constructed as mean differences: $v_W = \bar{x}_{\text{high warmth}} - \bar{x}_{\text{low warmth}}$ and $v_C = \bar{x}_{\text{high comp.}} - \bar{x}_{\text{low comp.}}$ All four models were run under strictly identical conditions (same stories, same pooling rule, same seed 20260527) to enable cross-architecture comparison.

PCA Denoising. The warmth and competence direction vectors share a substantial common component reflecting general narrative valence: stories portraying protagonists positively (whether warmly or competently) occupy a similar region of the residual stream, as do negatively valenced stories. To separate conceptual content from evaluative polarity, we follow the neutral-corpus denoising procedure of Sofroniew et al. (2026).

We construct a neutral corpus of 1,500 Wikipedia introduction paragraphs, length-matched to the concept stories (90–200 words) and filtered to exclude emotionally charged or violent content. Residual-stream activations are extracted at the same layer and pooling rule. Principal component analysis (PCA) is fitted on these neutral activations; the top components collectively explaining at least 50 per cent of neutral variance are then projected out of both direction vectors. For Gemma-3-12B the first principal component alone explains 56.1 per cent of neutral variance; projecting it out reduces the cosine between the warmth and competence vectors from 0.749 to 0.530. For Gemma-3-27B, 43 components are required to cross the 50 per cent threshold (50.2 per cent explained), reducing the inter-axis cosine from 0.708 to 0.487. The remaining post-denoising cosine (0.53 and 0.49, respectively) is consistent with the genuine warmth-competence correlation observed in human ratings ($r \approx 0.61$; Fiske et al. 2002), suggesting that the residual overlap reflects real conceptual co-variation rather than a denoising failure. The resulting denoised vectors are used for all causal steering experiments.

Probe Validation. We assess the validity of each direction vector using five complementary metrics. (i) *5-fold cross-validated accuracy*: a logistic regression classifier is trained on full residual-stream representations within each target contrast; accuracy is averaged across five stratified folds. (ii) *Topic-holdout accuracy*: a stricter variant using GroupKFold, where all stories from a given topic are assigned to the same fold, testing whether the probe generalises to entirely unseen situations rather than merely to unseen phrasings of familiar topics. (iii) *Cohen's d*: the standardised mean difference between high- and low-condition projections onto the direction vector, compared against an empirical null distribution of 1,000 random unit vectors. (iv) *Split-half cosine stability*: each condition's 50 stories are randomly split into halves of 25; a direction is computed from each half and the cosine between them is reported. (v) *Cross-axis classification*: using fold-local standardised projections, we test whether the warmth direction predicts competence labels and vice versa, quantifying the degree of shared evaluative content. Per-story Spearman rank correlations across model pairs are also computed to test whether the four models converge on a shared cross-architecture construct.

Causal Steering. To test whether the extracted directions causally influence model behaviour rather than merely correlating with it, we apply activation steering (Turner et al., 2023) via TransformerLens hooks at the probe layer. Steering strength is expressed relative to the mean residual-stream norm at that layer, ensuring comparability across models whose activation scales differ by orders of magnitude (see $\|\bar{h}\|$ in Tab. 1). We use a local regime of $\alpha \in \{-0.10, -0.05, 0, +0.05, +0.10\}$ times the mean norm; a broader range is used as a saturation diagnostic. For *concept-level* evaluation, the model answers "Is the protagonist high in warmth/competence? Answer only Yes or No" on held-out topics (40 training / 10 test, determined by topic index). The outcome is the next-token logit margin $\Delta = \text{logit}(\text{Yes}) - \text{logit}(\text{No})$. For *hiring-level* evaluation, the steering hooks are applied while the model processes resume-evaluation prompts, and the outcome is the callback logit margin. We test six directions: the dense target direction, a decoded Gemma Scope 2 SAE direction (Google DeepMind Language Model Interpretability Team, 2025), an axis-specific component, a shared component, the opposing concept axis, and a random orthogonal control. Uncertainty is estimated via paired topic bootstrap resampling.

Benchmark. We compare model callback disparities against the warmth and competence ratings and callback rates from Gallo et al. (2024), which provides empirical grounding in human correspondence studies.

Data. Hiring stimuli are built around the 282 applicant names and category-level signals from Gallo et al. (2024), paired with a fixed administrative assistant resume template to isolate the effect of the social signal.

Results

Warmth and Competence Are Linearly Encoded Across Model Families. A cross-validated logistic regression probe achieves 100% accuracy on both the warmth and competence contrasts across all four tested models, in both standard 5-fold cross-validation and the stricter topic-holdout variant where entire situational topics are withheld from training. The perfect topic-holdout accuracy indicates that the probe captures something that generalises across entirely new social situations, not merely unseen phrasings of familiar topics. The separation strengths, measured as Cohen's d relative to a null distribution of 1,000 random unit directions, are exceptional for all models (Fig. 3): for Gemma-3-12B ($d = 2.70/2.83$ for warmth/competence, $z = 3.9/3.7$), Gemma-3-27B ($d = 2.95/3.27$), Qwen3-14B ($d = 8.97/9.97$, $z = 14.1/14.6$), and Llama-3.1-8B ($d = 8.48/9.07$, $z = 15.0/15.1$), zero of 1,000 random directions exceeded the target direction in any model. The layer-sweep emergence curves (Fig. 3) confirm that both Gemma models reach their peak separation at approximately 65 per cent of depth, consistent with the probe-layer selection in Tab. 1. The split-half cosine stability of the Gemma-3-12B direction vectors (0.83 for warmth, 0.88 for competence) confirms that these directions are stable properties of the conditions rather than artefacts of specific stories. Critically, the four models do not merely separate high from low conditions in isolation: per-story Spearman rank correlations across model pairs reveal that they agree substantially on which specific stories are warm or cold ($\rho = 0.76$–0.98 for warmth across pairs; 0.78–0.99 for competence), indicating convergence on a shared cross-architecture construct rather than independent idiosyncratic representations. The cosine similarity between warmth and competence direction vectors prior to denoising (0.749 for Gemma-3-12B; 0.708 for Gemma-3-27B; 0.51–0.54 for Qwen3 and Llama) and the cross-axis classification accuracy (0.82–0.87 for Gemma; $\approx$1.00 for Qwen3/Llama) indicate that the two directions share a common evaluative component reflecting the positive-vs-negative framing of the stories. After PCA denoising, the Gemma inter-axis cosine drops to 0.530 (12B) and 0.487 (27B), while the post-denoising residual remains consistent with the genuine warmth-competence correlation reported in human perception studies ($r \approx 0.61$).

Warmth and Competence Directions Are Causally Functional. To move beyond representational description, we applied activation steering to the warmth and competence direction vectors in Gemma-3-12B and Gemma-3-27B on held-out concept judgement tasks. Before intervention, the model answered direct warmth/competence Yes/No questions with 100% accuracy on held-out topics, with baseline high–low logit-margin gaps of 16.14/14.10 (12B) and 25.93/23.64 (27B). Steering the dense target directions in the local regime produced positive, approximately linear causal effects in all four model-by-axis combinations (slopes: 12B warmth 27.79, $R^2 = 0.956$; 12B competence 15.11, $R^2 = 0.915$; 27B warmth 12.89, $R^2 = 0.990$; 27B competence 8.85, $R^2 = 0.826$). The steering effect does not support a clean two-module interpretation, however: the opposing concept axis also shifts judgements substantially in several conditions. Rather than treating this as a failure of specificity, we read it as a positive finding that connects to the structure of the human benchmark. Gallo et al. (2024) observe that warmth and competence ratings reduce to a dominant shared principal component that predicts callback rates; our steering results suggest that the same evaluative structure is represented inside the model, such that moving along either dimension also moves the shared positive-vs-negative appraisal. Cross-scale feature-profile correlations between matched Gemma-3-12B and Gemma-3-27B Gemma Scope 2 features (mean $r = 0.49$–0.66, exceeding a 500-permutation row-shuffle null at $p = .002$ for all five vector types) further indicate that the representational organisation is conserved across scale.

Concept-Level Steerability Differs Substantially Across Models. To compare how efficiently stored representations can be causally shifted, we extended the dense steering evaluation to all four models on held-out concept judgement tasks, normalising each model's peak effect ($\alpha = +0.10$) by its baseline high–low logit gap to obtain a *normalised steerability* score that removes the confound of differing baseline discrimination strengths. The resulting ranking (Fig. 4) shows Gemma-3-12B as the most steerable model (warmth = 0.236, competence = 0.140), followed by Qwen3-14B (warmth = 0.125, competence = 0.103), Gemma-3-27B (warmth = 0.040, competence = 0.009), and Llama-3.1-8B (warmth = 0.029, competence = 0.024). The substantially lower normalised steerability of Gemma-3-27B and Llama-3.1-8B relative to their probe accuracy indicates that these models organise the warmth/competence concept in a form less directly accessible to the mean-difference direction vector, despite the fact that the direction still perfectly separates high from low stories. This steerability ordering acquires an unexpected character when connected to the hiring-level mediation results described below: the most concept-steerable model (Gemma-3-12B, warmth steerability = 0.236) shows no statistically significant indirect effect in the hiring-outcome mediation, while the model with the lowest warmth steerability (Llama-3.1-8B, 0.029) shows the strongest bootstrap mediation indirect effect (IE = +0.190, 95% CI excluding zero). This reversal indicates that the capacity to shift a model's concept representations via linear activation addition does not map monotonically onto the causal role those representations play in downstream hiring decisions.

The Model's Internal Probe Scores Track Human Social Perceptions Unevenly Across Architectures. Before testing causal effects on hiring outcomes, we establish the degree to which each model's internal probe scores reflect human social perception ratings from Gallo et al. (2024). For each of the 282 applicant names, we extract the model's residual-stream activation at the probe layer and project it onto the warmth and competence direction vectors. The Gemma family shows positive alignment with human ratings: Spearman $\rho = +0.366$ (warmth) and $\rho = +0.239$ (competence) for Gemma-3-12B, rising to $\rho = +0.396$ and $\rho = +0.272$ for Gemma-3-27B (all $p < .001$). This indicates that names associated with groups perceived as warmer or more competent by humans are also represented as such internally in the Gemma family, and that scale slightly improves this alignment. The two non-Gemma models show a strikingly different pattern. Llama-3.1-8B anti-aligns on warmth ($\rho = -0.300$, $p < .001$) with a non-significant competence correlation ($\rho = -0.063$, $p = .29$). Qwen3-14B also anti-aligns on warmth ($\rho = -0.193$, $p = .001$) while aligning strongly

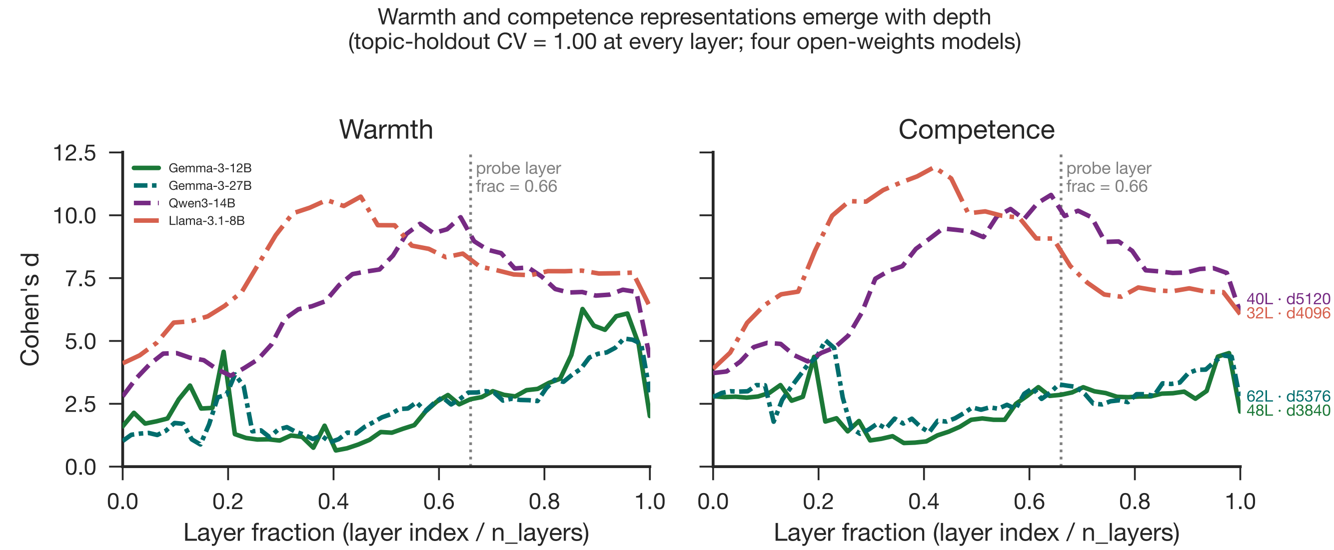

Figure 3: Warmth and competence directions emerge robustly across depth. Layer-wise concept separation and warmth-competence geometry across the four open-weights models.

on competence ($\rho = +0.465$, $p < .001$). This warmth anti-alignment is not the result of arbitrary noise: the cross-model story-level agreement between Gemma and Llama ($\rho = 0.768$ for warmth) and Gemma and Qwen ($\rho = 0.760$ for warmth) confirms that all four models agree on the relative warmth ordering of stories. The anti-alignment therefore implies that the direction these models share in activation space is systematically inverted relative to the human warmth scale: what the models represent as the "high warmth" pole corresponds to the low end of human warmth ratings for these two architectures. We attribute this inversion to instruction tuning and reinforcement learning from human feedback: the fine-tuning objective may have restructured the warmth axis such that social signals stereotypically associated with high warmth in raw pretraining text are now suppressed or inverted in the aligned model's representations. At Gemma-3-12B the internal warmth and competence probe scores predict the model's baseline callback margin in the same direction as human ratings ($\rho = +0.139$, $p = .084$; $\rho = +0.145$, $p = .067$), while at Gemma-3-27B this association reverses: names the model represents as warmer or more competent internally receive lower callback margins ($\rho = -0.160$ and -0.156, respectively), opposite to the 12B pattern. This reversal at 27B coincides with a markedly different baseline response regime: the 27B model recommends callbacks at an implied $P(\text{Yes}) = 0.760$ versus 0.452 at 12B.

Steering Warmth and Competence Causally Shifts Callback Recommendations. To test whether directly manipulating warmth and competence representations moves the model's hiring-callback decision, we applied activation steering during resume evaluation across 60 applicant names drawn from the Gallo et al. (2024) rated name set. At Gemma-3-12B the baseline callback margin was -0.194 ($SD = 0.130$), corresponding to an implied $P(\text{Yes}) = 0.452$. In the local steering regime, increasing warmth at $\alpha = +0.05$ produced a mean shift of $+1.177$ (SE $= 0.018$) and at $\alpha = +0.10$ a shift of $+2.369$ (SE $= 0.017$), a large and approximately linear positive effect on the callback margin. The competence direction at 12B shows a more complex pattern: negative steering suppresses the margin ($\alpha = -0.05 \rightarrow -1.515$) while positive steering produces only modest increases ($\alpha = +0.10 \rightarrow +0.231$), indicating that the competence direction does not have a symmetric causal effect on hiring decisions at this scale. Replicating the causal evaluation at Gemma-3-27B reveals a qualitatively different pattern for warmth. At $\alpha = +0.05$ the mean callback shift is $+1.973$, comparable to the 12B effect, but at $\alpha = +0.10$ the effect collapses to -2.658: an increase in steering strength reverses rather than amplifies the direction of effect. This non-monotone response rules out simple ceiling saturation as an explanation, because a ceiling effect would produce a plateau rather than a sign reversal, and the 27B baseline rate ($P(\text{Yes}) = 0.760$) leaves substantial headroom for upward shifts. The warmth representation in Gemma-3-27B therefore has a narrow window within which causal steering produces the expected effect; beyond that window, the representational intervention disrupts the decision process in a non-linear way. The full four-model dose-response curves and signal-vs-control comparisons across all six steering directions are shown in Fig. 5 and Fig. 6.

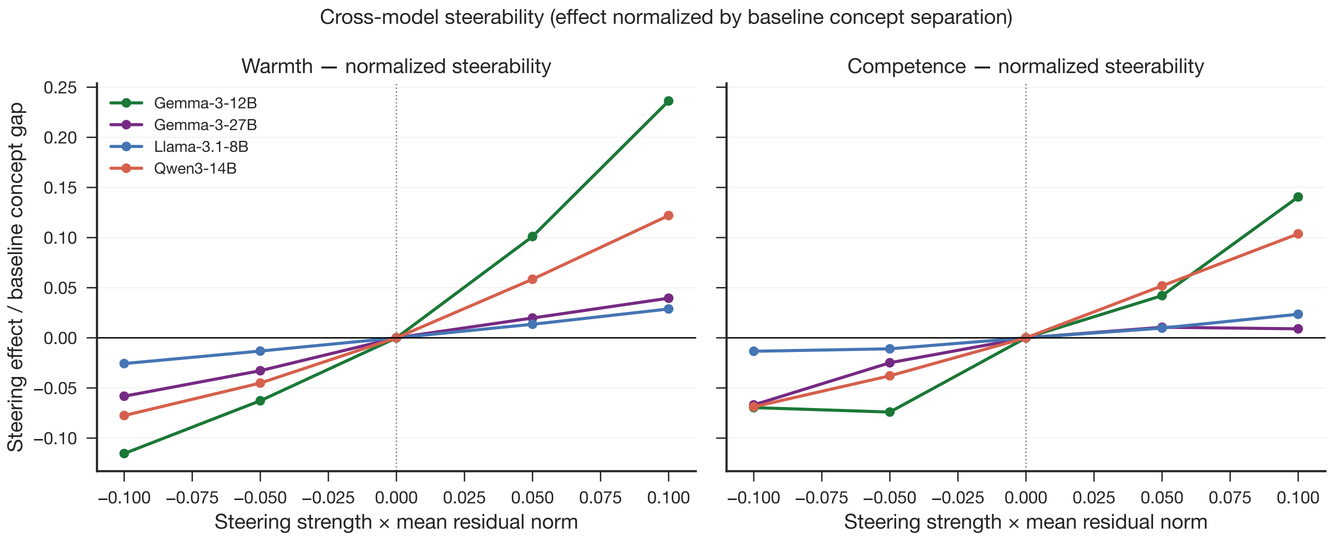

Figure 4: Concept-level steerability differs across architectures. Dense steering effects normalised by each model's baseline high–low concept gap, allowing steerability to be compared across models with different logit scales.

Model Callback Disparities Versus the Human Benchmark. To assess demographic parity in model hiring recommendations relative to human correspondence-study outcomes, we joined the 282-name stimulus set to the Gallo et al. (2024) published callback benchmark on first-name match (149 of 282 names matched; 9 Black Female, 9 Black Male, 84 White Female, 47 White Male). The human benchmark shows the classical audit-study pattern: White names receive substantially higher callback rates than Black names (race gap $= -0.085$ in favour of White), and male names receive slightly higher rates than female names (gender gap $= -0.037$ in favour of Male). At Gemma-3-27B the race gap is reversed relative to the human benchmark and large in magnitude: Black-signalling names receive callback margins $+0.486$ raw logit units above White-signalling names, equal to $+1.18$ within-model standard deviations (Fig. 7). The gender gap at 27B is -0.211 raw logit units (-0.51 SD), reproducing the human direction (Female $<$ Male) though at greater magnitude than the human reference. These results indicate that the larger Gemma model has overcorrected the human racial discrimination pattern: it gives a large systematic advantage to Black-signalling names, a direction opposite to the empirical record, while preserving the gender gap direction. This pattern is consistent with reinforcement learning from human feedback having corrected racial associations so strongly that the resulting calibration overshoots the empirical distribution of human outcomes by more than a standard deviation. At Gemma-3-12B the callback margin variance is too low (SD $= 0.14$, seven unique values on the 0.125 logit grid) to detect group-level disparities reliably; this model's R4 results are disclosed in the Limitations. At the name level, an exploratory ordinary least-squares regression predicting callback margin from the model's own probe scores ($n = 149$ matched names) shows warmth aligned with the causal direction at 12B (Pearson $r = +0.376$, $p < .001$) and reversed at 27B ($r = -0.266$, $p = .001$; $R^2 = 0.088$), consistent with the reversed baseline association documented above. Disparity results for Llama-3.1-8B and Qwen3-14B await completion of cluster reruns and are not reported here.

Discussion and Conclusion

That large language models trained on human-generated text would encode warmth and competence as internally recoverable representations is, in one sense, not surprising: these models distil patterns from the same texts in which warmth and competence shape social perception. What the present results establish empirically is that this encoding is linear, geometrically stable, causally functional at the concept level, and shared across architecturally distinct models from independent laboratories. Each of these properties matters: linearity makes the representation accessible to lightweight probes and steering interventions; causal functionality means the representation participates in explicit judgements about warmth and competence; and cross-architecture convergence suggests the finding reflects a property of training data and the objectives of language modelling rather than a quirk of any single design.

The scale comparison between Gemma-3-12B and Gemma-3-27B complicates this picture in an instructive way. Both models encode warmth and competence with comparable precision, and the 27B model's probe scores

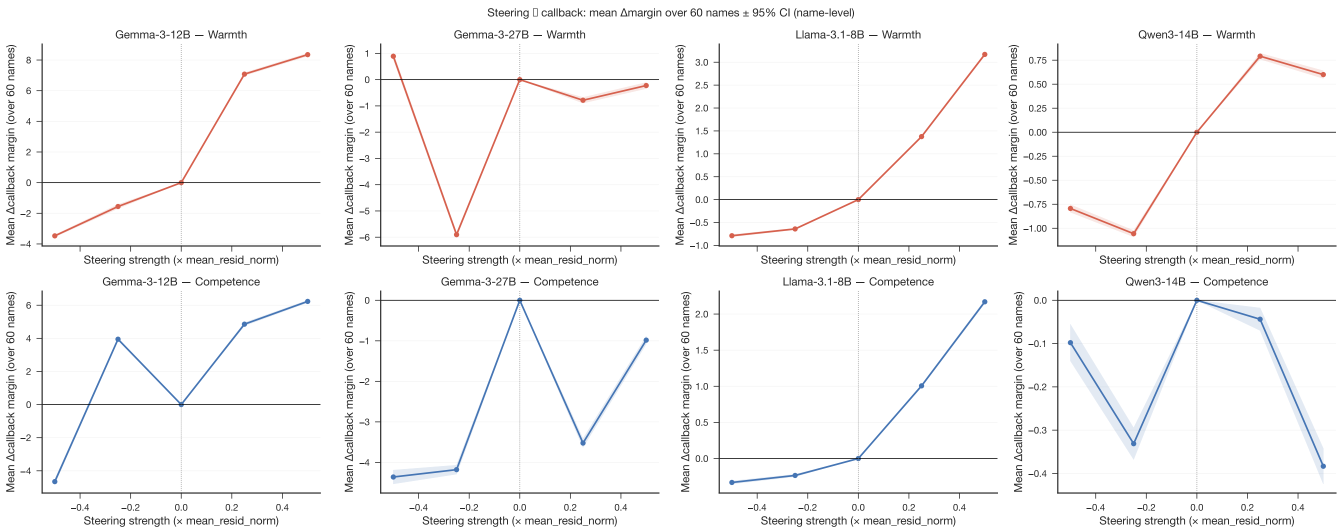

Figure 5: Activation steering shifts hiring callback recommendations. Callback-margin responses to warmth and competence steering in the hiring evaluation setting.

correlate slightly more strongly with human warmth and competence ratings. Yet the causal chain from concept representation to hiring decision differs sharply between them. At 12B, increasing the warmth representation produces a large, approximately linear increase in callback inclination (up to +2.37 logit units at $\alpha = +0.10$). At 27B, the equivalent intervention produces a modest positive shift at low strength (+1.97 at $\alpha = +0.05$) that collapses to a large negative shift at slightly higher strength (−2.66 at $\alpha = +0.10$). This non-monotone response is not saturation: the 27B baseline already sits at $P(\text{Yes}) = 0.76$, leaving room for upward movement, and the response is a sign reversal rather than a plateau. A more plausible account is that the 27B decision layer is organised by multiple interacting representations; the concept direction, when amplified beyond a narrow window, begins to interfere with those interactions rather than reinforcing them.

A related puzzle emerges from the cross-model steerability analysis. Gemma-3-12B is the most concept-steerable model (normalised warmth steerability = 0.236), yet it shows no significant hiring-level mediation path. Llama-3.1-8B is the least concept-steerable model (normalised warmth steerability = 0.029) and shows the strongest hiring-level indirect effect (bootstrap IE = +0.190). This steerability paradox illustrates that the capacity to shift a model's representations via linear probing does not imply that those representations causally govern the downstream decision. Models in which representations are most cleanly isolable may be precisely those in which decision processes operate through parallel mechanisms that bypass the steering target.

The warmth anti-alignment observed in Llama-3.1-8B and Qwen3-14B adds a further layer of complexity. Despite high cross-model story agreement (per-story Spearman $\rho \approx 0.76$–0.77 between Gemma and each of the other models), the direction that Llama and Qwen organise as "high warmth" in activation space is negatively correlated with human warmth ratings ($\rho = -0.300$ and -0.193 respectively). The models converge on the same relative ordering of stories yet that shared ordering is inverted relative to the human scale. We interpret this as evidence that instruction tuning has restructured the warmth axis in these architectures: reinforcement learning from human feedback may have adjusted representations associated with stereotypically warm social groups in a direction that happens to invert the relationship to the original pretraining warmth axis.

The demographic disparity results at Gemma-3-27B illustrate the practical consequence of this representational reorganisation. The model strongly advantages Black-signalling names relative to White-signalling names (+1.18 SD), directly reversing the classical correspondence-study finding. The gender gap direction, by contrast, reproduces the human pattern. The asymmetry between race and gender suggests that instruction tuning has applied differential corrections to different demographic axes, producing a model that is not more equitable overall but differently inequitable: the racial hierarchy is reversed rather than eliminated, and at substantially larger magnitude than the original human disparity. This finding points to a fundamental limitation of outcome-level bias mitigation: correcting behavioral outputs without understanding the representational substrate can shift rather than remove the underlying imbalance.

Together, the results suggest that LLMs have internalised a structure of social perception broadly analogous to the SCM, emerging from training on human-

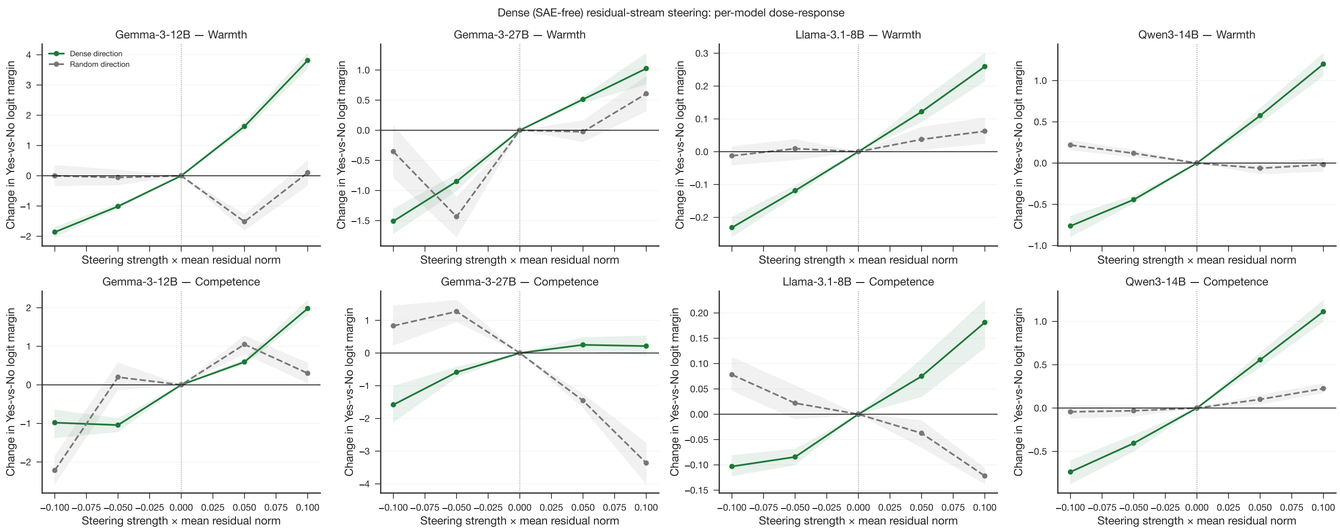

Figure 6: Dense steering dose-response across models. Concept-level dose-response curves for warmth and competence directions across the four open-weights models.

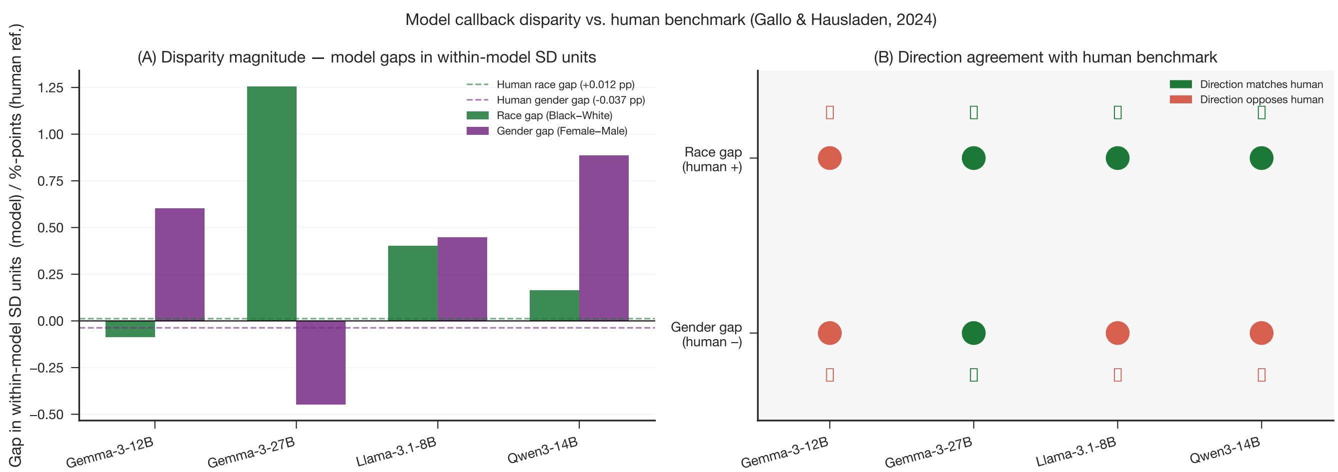

Figure 7: Model callback disparities relative to the human benchmark. Race and gender callback-margin gaps for the model evaluations compared with the correspondence-study benchmark.

generated text rather than through any explicit design choice. The connection from that internal structure to actual hiring decisions is not, however, simple or consistent across models. This finding points to a methodological implication for bias auditing: probing internal representations and measuring output disparities are complementary but not interchangeable, and a model that performs well on one measure may not perform well on the other. It also opens a path to addressing the problem at the source by intervening on the representation rather than filtering outputs after the fact, and provides a methodological template for connecting the computational tools of mechanistic interpretability to the empirical tradition of correspondence studies in the social sciences.

Limitations. The concept story corpus was generated by a large language model, raising concerns about circularity in a study that probes language model representations. We note, however, that the generating model (Claude, Anthropic) and the probed models (Gemma, Qwen, Llama) come from different organisations and architectures, so the warmth/competence contrasts are not trivially reproduced by a shared generator. Nevertheless, the possibility that AI-generated stimuli share stylistic properties that inflate probe accuracy across model families cannot be fully excluded; replication with human-authored or template-based stimuli is desirable. The current corpus of 200 stories is a pilot that covers 50 topics; the planned expansion to 400 stories will increase per-condition sample size and topic diversity. The probe-to-human validation correlations are moderate (warmth ρ = +0.366 and +0.396 for 12B and 27B; competence ρ = +0.239 and +0.272), indicating that the model's internal probes capture a component of human social perception rather than replicating it faithfully. The warmth anti-alignment in Llama-3.1-8B ($\rho = -0.300$) and Qwen3-14B ($\rho = -0.193$) means that for these models the mean-difference direction captures a construct that is systematically inverted relative to the human warmth scale; results for these models should be interpreted with this inversion in mind. The causal warmth steering effect at the hiring level is positive and strong at 12B but non-monotone and fragile at 27B, so the causal claim does not generalise across scale without qualification. Callback margins are quantised to a 0.125-unit grid due to bf16 precision in the model's output logits; this quantisation is inherent to bf16 inference and is not resolved by float32 subtraction. At Gemma-3-12B the callback margin standard deviation is 0.14 with only seven unique values, making group-level disparity estimates at 12B unreliable and the 12B R4 results should not be cited as empirical findings. At Gemma-3-27B the larger variance (SD = 0.41, 18 unique values) makes group gaps of approximately 0.5 SD detectable, but the grid structure introduces discrete steps in all reported gap magnitudes. Demographic disparity results for Llama-3.1-8B and Qwen3-14B are pending cluster reruns with the corrected float32 subtraction and are not reported here. The hiring prompt is simplified relative to real-world screening contexts (single role, fixed qualifications), and results may differ under richer evaluation settings or different job types. Finally, the architectural source of the elevated warmth/competence vector cosine observed in Gemma-3 relative to Qwen3 and Llama-3.1 across all network depths remains an open question.

Future Work. The immediate next steps are extending the R4 demographic disparity analysis to Llama-3.1-8B and Qwen3-14B once the cluster re-runs are complete, and expanding the hiring stimuli to additional job roles to test whether results generalise beyond administrative assistant. The SAE decomposition performed here on Gemma-3 can be extended to Llama-3.1 using the Llama Scope sparse autoencoders (He et al., 2024), enabling a model-family comparison at the feature level. Validating the concept probes against a human-authored warmth/competence dataset would address the circularity concern associated with AI-generated stimuli. On the social-science side, extending the hiring stimuli beyond names to category-level signals (age, disability, religion) would broaden the scope of the benchmark comparison and test whether the representational account generalises beyond race and gender. Translating these findings into practical debiasing interventions and evaluating their robustness to adversarial prompting is a natural next step for applied work.

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Supplementary Materials

Details on Methods

Additional Results